Q(1)	Q(2)	Q(3)	Q(4)	Q(5)	Q(6)	Q(7)	Q(8)
B	A	D	D	C	C	B	A

Q(1) If $y = \tan^{-1}(\ln x)$, then $\frac{dy}{dx}|_{x=1} =$

a) -1

b) 1

c) 2

d) 0

Q(2) If $y = (x)^{\sinh x}$, then $\frac{dy}{dx}|_{x=1} =$

a) $\sinh(1)$

b) $\cosh(1)$

c) $\sinh(1) + \cosh(1)$

d) 0

Q(3) $\int_0^{\pi} \tan^4 x \, dx =$

a) 2π

b) 3π

c) 4π

d) π

Q(4) $\lim_{x \to 0^+} \left(\ln(\sinh x) - \ln(\sin^{-1} x)\right) =$

a) $\ln 2$

b) e

c) 1

d) 0

Q(5) $\int x\sqrt{9 - x^2}\, dx =$

a) $27 \int \tan\theta \sec^3\theta \, d\theta$

b) $9 \int \sin\theta \cos^2\theta \, d\theta$

c) $27 \int \sin\theta \cos^2\theta \, d\theta$

d) $3 \int \sin\theta \cos^2\theta \, d\theta$

Q(6) $\int \sin^6 x \cos^3 x \, dx =$

a) $\frac{1}{9}\sin^9 x - \frac{1}{7}\sin^7 x + c$

b) $\frac{1}{6}\sin^6 x - \frac{1}{8}\sin^8 x + c$

c) $\frac{1}{7}\sin^7 x - \frac{1}{9}\sin^9 x + c$

d) $\frac{1}{5}\sin^5 x - \frac{1}{7}\sin^7 x + c$

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Q(7) $\int x.2^x\,dx =$

a) $\dfrac{x.2^x}{(\ln 2)^2} - \dfrac{2^x}{\ln 2} + c$

b) $\dfrac{x.2^x}{\ln 2} - \dfrac{2^x}{(\ln 2)^2} + c$

c) $\dfrac{2^x}{\ln 2} - \dfrac{2^x}{(\ln 2)^2} + c$

d) $\dfrac{2^x}{\ln 2} - \dfrac{x.2^x}{(\ln 2)^2} + c$

Q(8) $\tan^2(\sin^{-1} x) =$

a) $\dfrac{x^2}{1-x^2}$

b) $\dfrac{1-x^2}{x^2}$

c) $\dfrac{x}{\sqrt{1-x^2}}$

d) $\dfrac{x^2}{x^2-1}$

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For equestions (9) – (12): Write T (True) or F (False).

Q(9) If $\log_x(2) = 3$, then $x = \sqrt[3]{2}$. (T)

Q(10) If $(e)^{e^{2x-1}} = 5$, then $x = \dfrac{1+\ln(\ln 5)}{2}$. (T)

Q(11) If $\lim\limits_{x\to 0}\dfrac{a^x-1}{x}$ exists, then $a \neq 0$. (T)

Q(12) The equation $\cosh x = \sinh x$ has a solution. (F)

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Q1: $\frac{dy}{dx} = \frac{1}{(\ln^2 x + 1)\, x}$

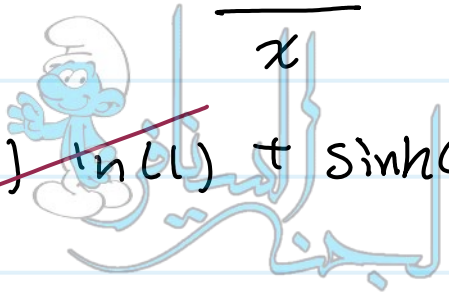
$$\left.\frac{dy}{dx}\right|_{x=1} = \frac{1}{(\ln 1 + 1)\, 1} = 1$$

Q2: $y = (\sinh(1))^{\sinh x} = 1$

$$\ln y = \sinh x \ln x$$

$$\frac{y'}{y} = \cosh x \ln x + \frac{\sinh x}{x}$$

$$\left.\frac{dy}{dx}\right|_{x=1} = \cosh(1) \ln(1) + \sinh(1)$$



Q3: By reduction formula:

$$I_n = \frac{\tan^{n-1} x}{n-1} - I_{n-2}$$

$$= \frac{\tan^3 x}{3}\Big]_0^{\pi} - \int_0^{\pi} \tan^2 x \, dx$$

$$= \int_0^{\pi} 1 - \sec^2 x \, dx = x\Big]_0^{\pi} - \tan x\Big]_0^{\pi} = \pi$$

Q4: $\ln\left(\lim_{x\to 0^+} \frac{\sinh x}{\sin^{-1} x}\right)$

using L·H: $\ln\left(\lim_{x\to 0^+} \cosh x \sqrt{1-x^2}\right)$

$= \ln(1) = 0$

Q5: $\int x\sqrt{9-x^2}\,dx \qquad \theta = \sin^{-1}\left(\frac{x}{3}\right)$

$= 3\int x(9-x^2)\,d\theta \qquad 3\sin\theta = x$

$= 27\int \sin\theta \cos^2\theta\, d\theta$

Q6: $\int \sin^6 x\,(1-\sin^2 x)\cos x\,dx \qquad u = \sin x$

$\frac{du}{\cos x} = dx$

$= \int u^6(1-u^2)\,du$

$= \int u^6 - u^8\,du = \frac{u^7}{7} - \frac{u^9}{9} + c$

$= \frac{\sin^7 x}{7} - \frac{\sin^9 x}{9} + c$

Q7:

u	dv
x	2^x
1	$2^x/\ln 2$
0	$2^x/(\ln 2)^2$

Q8: $u = \sin^{-1} x \longrightarrow 1 - x^2 = 1 - \sin^2 u = \cos^2 u$

$$\frac{1}{1-x^2} = \sec^2 u = \tan^2 u + 1$$

$$\tan^2 u = \frac{1}{1-x^2} - 1 = \frac{1-(1-x^2)}{1-x^2} = \frac{x^2}{1-x^2}$$

$$\tan u = \sqrt{\frac{x^2}{1-x^2}} \longrightarrow u = \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$$

$$\left(\tan\left(\tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)\right)\right)^2 = \left(\frac{x}{\sqrt{1-x^2}}\right)^2 = \frac{x^2}{1-x^2}$$

Q9: $\log_x(2) = 3 \longrightarrow \frac{\ln 2}{\ln x} \neq 3$

$$\ln x = \frac{1}{3}\ln 2 = \ln \sqrt[3]{2} \longrightarrow x = \sqrt[3]{2} \quad \checkmark$$

Q10: $e^{e^{2x-1}} = 5$

$e^{2x-1} = \ln 5 \longrightarrow 2x - 1 = \ln(\ln 5)$

$x = \dfrac{\ln(\ln 5) + 1}{2}$ ✓

Q11: if $a=0$: $\lim\limits_{x\to 0}\left(-\dfrac{1}{x}\right)$ = D.N.E

$\lim\limits_{x\to 0^+}\left(-\dfrac{1}{x}\right) = -\infty \neq \lim\limits_{x\to 0^-}\left(-\dfrac{1}{x}\right) = \infty$

but for example if $a>0$:

$\lim\limits_{x\to 0}\left(\dfrac{a^x-1}{x}\right)$ By L.H $= \lim\limits_{x\to 0}(a^x) = 1$

so the statement is true. ✓

Q12: $\dfrac{e^x + e^{-x}}{2} = \dfrac{e^x - e^{-x}}{2}$

$2e^{-x} = 0 \longrightarrow e^{-x} \neq 0$

S.S $= \{\} = \phi$ ✗